

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

12

Diagram NOT  
accurately drawn

$$9x + 21 + 2x + 21 = 150 \quad (\text{Perimeter of lot})$$

$$9x + 2x + 21 + 21 = 150$$

$$11x + 42 = 150$$

$$11x = 150 - 42$$

$$\frac{11x}{11} = \frac{108}{11}$$

$$x = 9.8$$

$$\text{Lot } = 6 \quad (9 \times 10 + 21 = 111)$$

$$\text{Perimeter} = \frac{\text{Lot} \times 4}{11}$$

$$P = 171 = \frac{4}{11} \times 11$$

$$\frac{4}{11} \times 11 = 4$$

$$n = \frac{4}{11}$$

(Total for Question 12 is 4 marks)

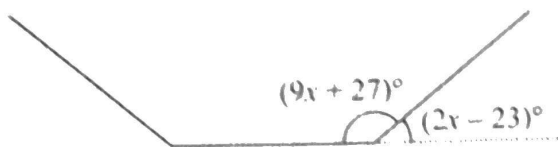


Diagram NOT  
accurately drawn

The diagram shows part of a regular  $n$ -sided polygon with

an interior angle of  $(9x + 27)^\circ$

an exterior angle of  $(2x - 23)^\circ$

Work out the value of  $n$

$$9x + 27 + 2x - 23 = 180 \quad (\text{as in straight line})$$

$$9x + 2x + 27 - 23 = 180$$

$$11x + 4 = 180$$

$$11x = 180 - 4$$

$$\frac{11x}{11} = \frac{176}{11}$$

$$x = 16$$

$$\therefore \text{Interior } \angle = 9 \times 16 + 27 = 171$$

$$1 \text{ Interior } \angle = \frac{\text{Int } \angle}{n}$$

$$n \times 171 = \frac{360}{n} \times n$$

$$\frac{360}{171} = \frac{360}{n}$$

$$n = \frac{360}{171}$$

(Total for Question 12 is 4 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

